Integral Cheat Sheet (Immediate Integrals + Identities + Substitutions)

No sec / csc notation: use $1/\cos$ and $1/\sin$ • Includes cot

0) Core rules (always valid)

$$\int(\alpha f+\beta g)\,dx=\alpha\int f\,dx+\beta\int g\,dx,\qquad \int f'(x)\,dx=f(x)+C$$

$$\int f(x)\,f'(x)\,dx=\frac{1}{2}f(x)^2+C,\qquad \int\frac{f'(x)}{f(x)}\,dx=\ln|f(x)|+C$$

$$\int(f(x))^n f'(x)\,dx=\frac{(f(x))^{n+1}}{n+1}+C\quad(n\neq-1)$$

1) Power / algebraic (immediate)

$$\int x^n\,dx=\frac{x^{n+1}}{n+1}+C\quad(n\neq-1),\qquad \int\frac{dx}{x}=\ln|x|+C$$

$$\int(ax+b)^n\,dx=\frac{(ax+b)^{n+1}}{a(n+1)}+C\quad(n\neq-1)$$

$$\int\frac{dx}{ax+b}=\frac{1}{a}\ln|ax+b|+C$$

2) Exponential (immediate)

$$\int e^x\,dx=e^x+C,\qquad \int e^{kx}\,dx=\frac{1}{k}e^{kx}+C\ (k\neq0)$$

$$\int a^x\,dx=\frac{a^x}{\ln a}+C\quad(a>0,\ a\neq1),\qquad \int a^{kx}\,dx=\frac{a^{kx}}{k\ln a}+C$$

3) Logarithms (immediate / common)

$$\int\ln x\,dx=x\ln x-x+C\quad(x>0)$$

$$\int\frac{\ln x}{x}\,dx=\frac{(\ln x)^2}{2}+C\quad(x>0)$$

4) Trigonometric integrals (no $\sec/\csc$)

$$\int\sin x\,dx=-\cos x+C,\qquad \int\cos x\,dx=\sin x+C$$

$$\int\tan x\,dx=-\ln|\cos x|+C,\qquad \int\cot x\,dx=\ln|\sin x|+C$$

$$\int\frac{dx}{\cos^2 x}=\tan x+C,\qquad \int\frac{dx}{\sin^2 x}=-\cot x+C$$

$$\int\frac{\sin x}{\cos^2 x}\,dx=\frac{1}{\cos x}+C,\qquad \int\frac{\cos x}{\sin^2 x}\,dx=-\frac{1}{\sin x}+C$$

$$\int\frac{dx}{\cos x}=\ln\left|\frac{1+\sin x}{\cos x}\right|+C,\qquad \int\frac{dx}{\sin x}=\ln\left|\tan\frac{x}{2}\right|+C$$

5) Standard inverse-trig / radical forms

$$\int\frac{dx}{a^2+x^2}=\frac{1}{a}\arctan\Big(\frac{x}{a}\Big)+C\quad(a>0)$$

$$\int\frac{dx}{\sqrt{a^2-x^2}}=\arcsin\Big(\frac{x}{a}\Big)+C\quad(a>0)$$

$$\int\frac{dx}{\sqrt{a^2+x^2}}=\ln\left|x+\sqrt{a^2+x^2}\right|+C\quad(a>0)$$

$$\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|+C\quad(a>0)$$

$$\int\frac{dx}{\sqrt{x^2-a^2}}=\ln\left|x+\sqrt{x^2-a^2}\right|+C\quad(a>0)$$

6) Basic trigonometric identities

$$\sin^2 x+\cos^2 x=1,\qquad 1+\tan^2 x=\frac{1}{\cos^2 x},\qquad 1+\cot^2 x=\frac{1}{\sin^2 x}$$

$$\sin(2x)=2\sin x\cos x$$

$$\cos(2x)=\cos^2 x-\sin^2 x=2\cos^2 x-1=1-2\sin^2 x$$

$$\tan(2x)=\frac{2\tan x}{1-\tan^2 x}$$

$$\sin^2 x=\frac{1-\cos 2x}{2},\qquad \cos^2 x=\frac{1+\cos 2x}{2}$$

7) Trigonometric substitutions (for radicals)

$$\sqrt{a^2-x^2}:\ x=a\sin\theta,\ dx=a\cos\theta\,d\theta$$

$$\sqrt{a^2+x^2}:\ x=a\tan\theta,\ dx=a\frac{1}{\cos^2\theta}\,d\theta$$

$$\sqrt{x^2-a^2}:\ x=\frac{a}{\cos\theta},\ dx=a\frac{\sin\theta}{\cos^2\theta}\,d\theta$$

8) Universal trigonometric substitution (Weierstrass)

$$t=\tan\left(\frac{x}{2}\right),\qquad \sin x=\frac{2t}{1+t^2},\qquad \cos x=\frac{1-t^2}{1+t^2},\qquad dx=\frac{2}{1+}$$

Use when integrand is a rational function of $\sin x$ and $\cos x$.